

file to use. In case you run PhotoRec directly you'll get prompt which asks for which device to use. Select the device on which you'd like to attempt recovery. Followed by the partition table type, in most cases it's Intel, unless you have changed your partition table type. After this you'll be asked to select the partition on which you'd like to run PhotoRec. Select the filesystem

in free space only or the whole disk. After this you'll be asked for the output directory where you want the recovered files to be places. That's it PhotoRec will have started running and copying recovered files to the

## Test Disk

TestDisk is primarily designed to help recover lost partitions and/or make non-booting disks bootable. It also has an option

```
PhotoRec 6.11, Data Recovery Utility, April 2009
Christophe GRENIER <grenier@cgsecurity.org>
http://www.cgsecurity.org

PhotoRec is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media (use Arrow keys, then press Enter):
Disk /dev/sda - 80 GB / 74 GiB (RO) - ATA ST380810AS
Disk /dev/sdb - 160 GB / 149 GiB (RO) - WD 1600BMV External

[Proceed] [Quit]

Note:
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has incorrect size, check HD jumper settings, BIOS
detection, and install the latest OS patches and disk drivers.
```

Select the appropriate device, you'll not get this screen if you've opened PhotoRec specifying an image

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Disk /dev/sdb - 160 GB / 149 GiB (RO) - WD 1600BMV

Partition      Start      End
No partition    0 0 1 19457 80 63
1 * FAT32 LBA    0 1 1 19456 254 63
```

Select the partition from which you want to recover the data

```
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1 * FAT32 LBA    0 1 1 19456 254 63 312576642 [LIVE]

Please choose if all space need to be analysed:
[Free] Scan for file from FAT32 unallocated space only
[Whole] Extract files from whole partition
```

PhotoRec provides an option to either scan the whole disk or just the free area on the partition

type when asked. This is followed by a prompt which asks you whether to search for files

to recover files. The interface is much like Photorec. They're a part of the same package, Test-

```
kmaster@adsl:/src/testdisk
TestDisk 6.11-WIP, Data Recovery Utility, September 2008
Christophe GRENIER <grenier@cgsecurity.org>
http://www.cgsecurity.org

Disk /data/data_for_testdisk/ntfs-hdb1-img.dd - 98 MB / 94 MiB  C/S 12 253 63

[Analyse] Analyse current partition structure and search for lost partitions
[Advanced] Filesystem utility
[Geometry] Change disk geometry
[Options] Modify options
[Quit] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.
```

'Undelete' option in TestDisk is available under the Advanced menu

```
kmaster@adsl:/src/testdisk
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P NTFS 0 0 1 11 253 63 192717 [New Volume]

Deleted files
-----
0 0 660870 22-Jul-2008 20:13 /moodle/enrol/assessm.en.2.1.pdf
0 0 1714 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/COPYING
0 0 2132 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/Changelog
0 0 12451 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/LICENCE
0 0 1864 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/database.txt
0 0 365 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/metasploit2/.svn/all-wcprops
0 0 493 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/metasploit2/.svn/entries
0 0 2 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/metasploit2/.svn/format
0 0 124 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/metasploit2/.svn/text-base/README.svn-bu
0 0 8291 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/metasploit2/.svn/text-base/exploits.txt
0 0 14545 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/metasploit2/exploits.txt
0 0 111 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/samples/.svn/all-wcprops
0 0 285 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/samples/.svn/entries
```

Select the files you want to recover from the list provided

```
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0 0 365 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/metasploit2/.svn/all-wcprops
0 0 493 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/metasploit2/.svn/entries
0 0 2 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/metasploit2/.svn/format
0 0 124 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/metasploit2/.svn/text-base/README.svn-bu
0 0 8291 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/metasploit2/.svn/text-base/exploits.txt
0 0 14545 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/metasploit2/exploits.txt
0 0 111 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/samples/.svn/all-wcprops
0 0 285 22-Jul-2008 20:07 /moodle/lib/gisip/Documents/samples/.svn/entries
```

TestDisk will copy the recovered file to the output directory, it will not delete the original data

Disk is also bundled with this months Alpha DVD. Select the appropriate hard disk, partition table type, partition and file system. This will bring you to the main menu. In the menu go to 'Advanced', select the partition that was holding the lost files and choose 'Undelete'. After this TestDisk scans MFT entries for deleted files. Go through the files that Test

Disk just found out select the files you want by pressing c. Once done proceed to select the output folder and you're done, TestDisk will copy those recovered file to the destination.

## NTFSUndelete

Windows 7 and Vista both routinely put a "recovery partition" at the front of drives now even as fresh

```
jait@jait:~$ sudo ntfsundelete /dev/sdb1

Inode  Flags  %age  Date  Size  Filename
-----
16  F...  0%  2011-12-24  0 <none>
17  F...  0%  2011-12-24  0 <none>
18  F...  0%  2011-12-24  0 <none>
19  F...  0%  2011-12-24  0 <none>
20  F...  0%  2011-12-24  0 <none>
21  F...  0%  2011-12-24  0 <none>
22  F...  0%  2011-12-24  0 <none>
23  F...  0%  2011-12-24  0 <none>
```

Files with potentially recoverable content: 0  
Use NTFS Undelete to recover data from a Windows partition